Faculté : Sciences Informatique

Nom & Prénom : Louis Dochlie

TD N°3 – Reseaux

Niveau : L3

Date : Le/4/8/26

L’objectif de ce TD est de :

1. Apprendre à créer et gérer plusieurs VLANs pour isoler les différents départements (Administration, Commercial, Technique,

Invités, DMZ).

2. Comprendre le routage inter-VLAN à l’aide de routeurs et de sous-interfaces.

3. Identifier les avantages de la segmentation pour la sécurité.

4. Configurer des mots de passe forts pour l’accès console et VTY.

5. Activer SSH sur les routeurs pour un accès sécurisé à distance.

6. Appliquer la port-security sur les switchs pour limiter les adresses MAC autorisées sur chaque port.

7. Créer des ACL étendues pour contrôler l’accès entre VLANs (ex. : VLAN invités ne peuvent pas accéder à la DMZ ou aux

ressources sensibles).

8. Filtrer le trafic réseau de façon sélective (protocole, IP source/destination).

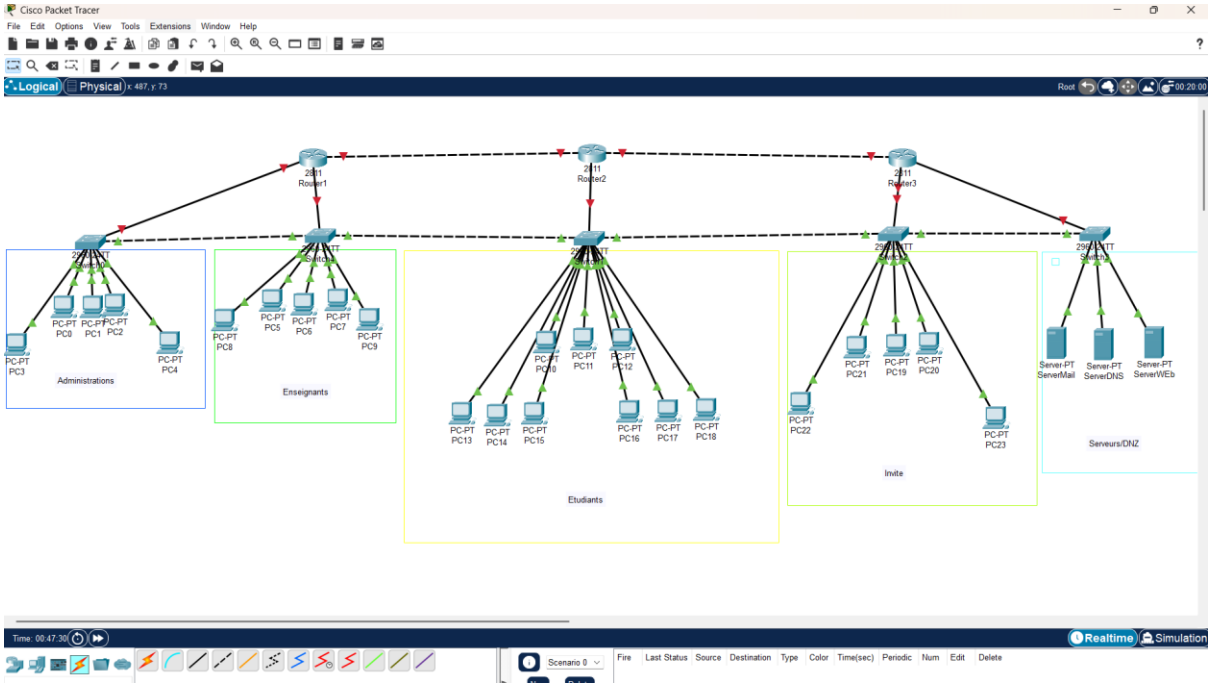
9. Simulation de scénarios d’attaque et test

1. Créer la topologie réseau

Placer 3 routeurs, 5 switches et 25 PC sur Packet Tracer.

Organiser les PCs selon leur rôle : administration, enseignants, étudiants, invités.

Connecter les switches aux routeurs et entre eux si nécessaire.



2. Configurer les VLANs

Créer des VLANs pour chaque département ou service :

VLAN 10 : Administration

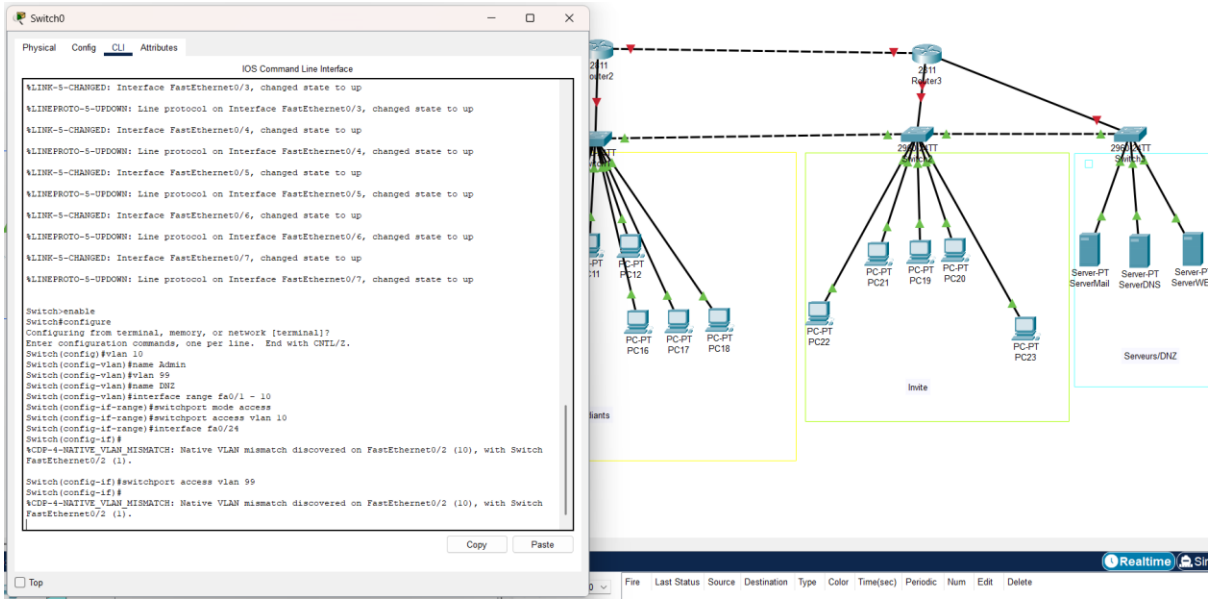
VLAN 20 : Enseignants

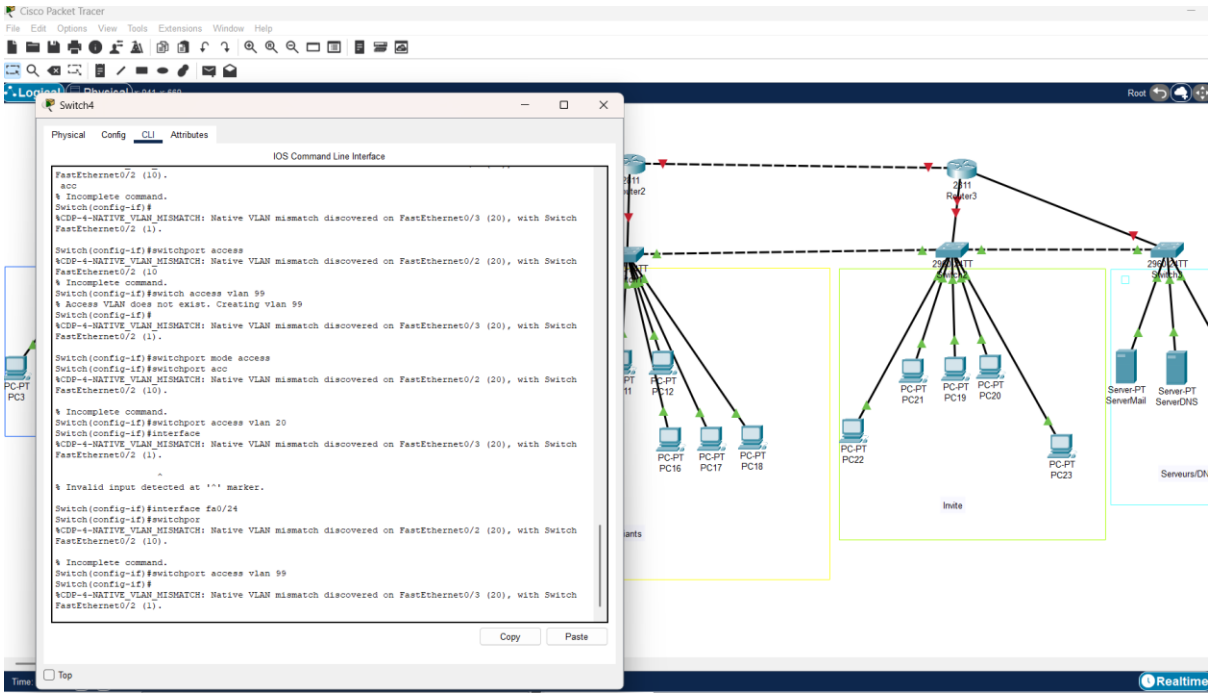
VLAN 30 : Étudiants

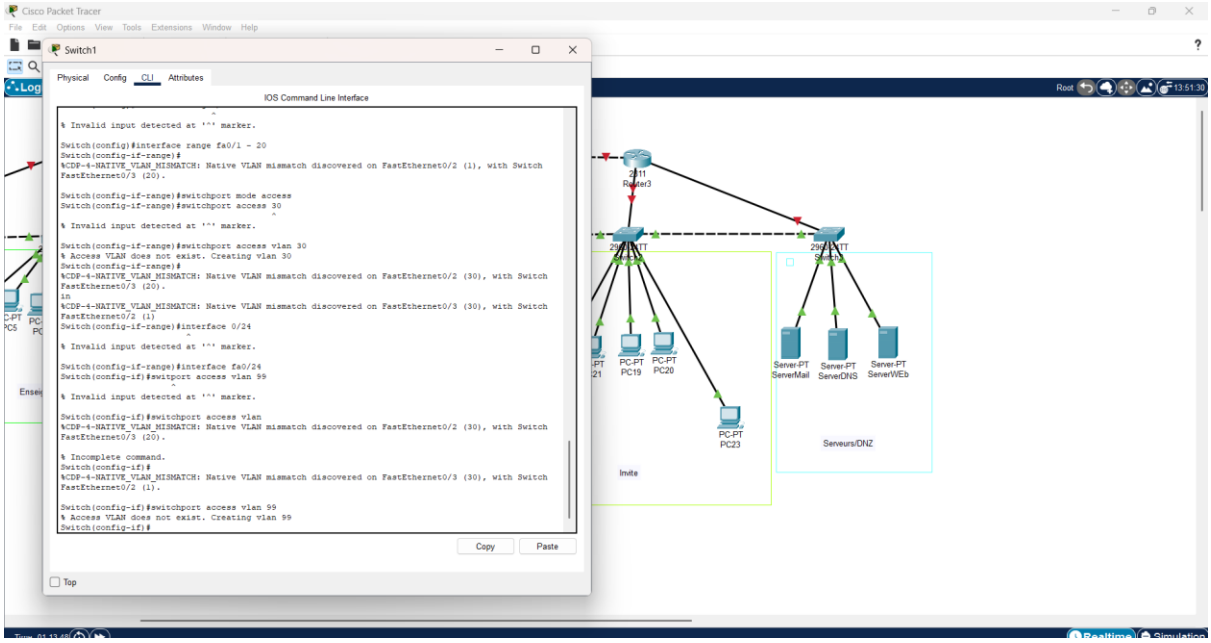
VLAN 40 : WiFi invités

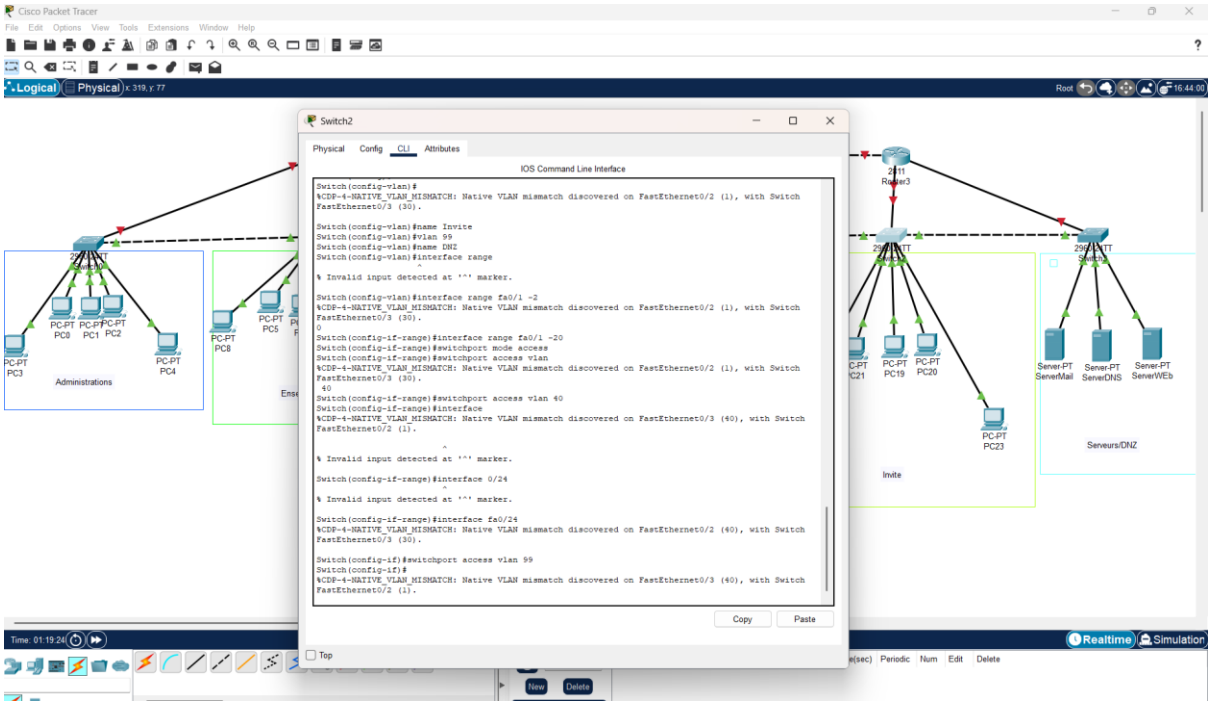
VLAN 99 : Serveurs / DMZ

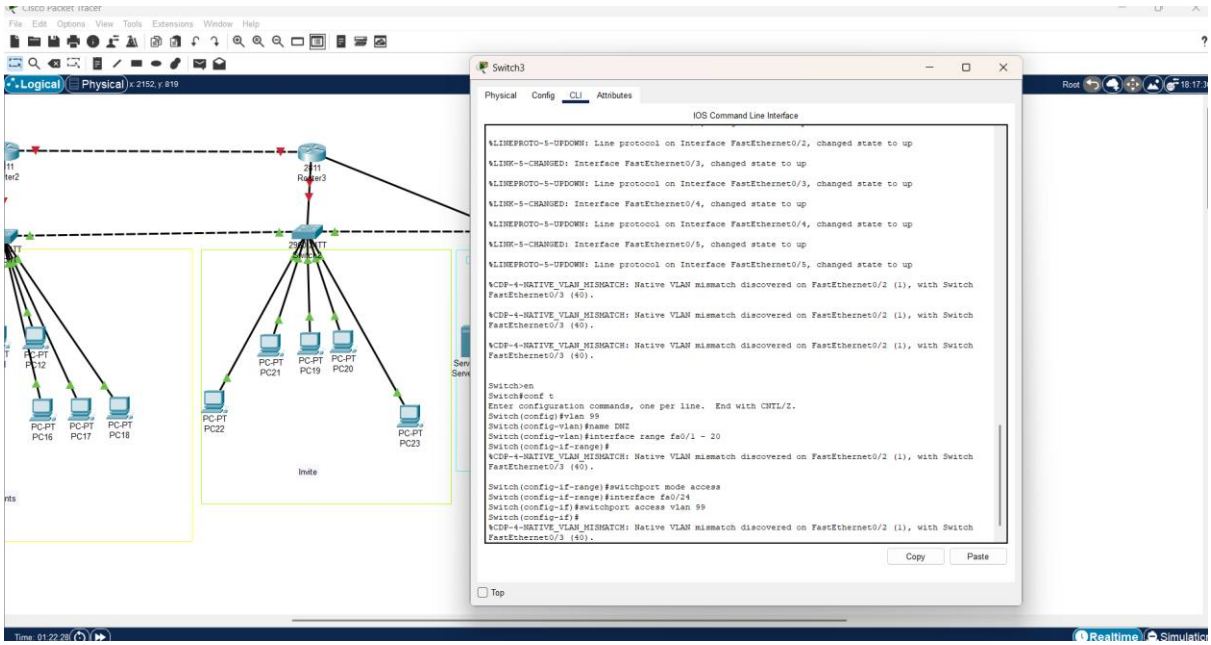
Assigner les ports des switches à leurs VLANs respectifs.


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
Switch>enable
Switch#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Admin
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch FastEthernet0/2 (1).
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch FastEthernet0/2 (1).
Copy
Paste
Top
Router3
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
PC-PT PC23
Invite
Server-PT ServerMail
Server-PT ServerDNS
Serveurs/DNZ
PC-PT PC16
PC-PT PC17
PC-PT PC18
Realtime
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Root
Switch4
Physical
Config
CLI
Attributes
IOS Command Line Interface
FastEthernet0/2 (10).
acc
% Incomplete command.
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (20), with Switch FastEthernet0/2 (1).
Switch(config-if)#switchport access
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with Switch FastEthernet0/2 (10
% Incomplete command.
Switch(config-if)#switch access vlan 99
% Access VLAN does not exist. Creating vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (20), with Switch FastEthernet0/2 (1).
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acc
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with Switch FastEthernet0/2 (10).
% Incomplete command.
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (20), with Switch FastEthernet0/2 (1).
% Invalid input detected at '^' marker.
Switch(config-if)#interface fa0/24
Switch(config-if)#switchpor
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (20), with Switch FastEthernet0/2 (10).
% Incomplete command.
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (20), with Switch FastEthernet0/2 (1).
Copy
Paste
Top
PC-PT PC3
Router3
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
PC-PT PC23
Invite
Server-PT ServerMail
Server-PT ServerDNS
PC-PT PC16
PC-PT PC17
PC-PT PC18
Realtime


Cisco Packet Tracer
Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
% Invalid input detected at '^' marker.
Switch(config)#interface range fa0/1 - 20
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with Switch FastEthernet0/3 (20).
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access 30
% Invalid input detected at '^' marker.
Switch(config-if-range)#switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with Switch FastEthernet0/3 (20).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (30), with Switch FastEthernet0/2 (1)
Switch(config-if-range)#interface 0/24
% Invalid input detected at '^' marker.
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switport access vlan 99
% Invalid input detected at '^' marker.
Switch(config-if)#switchport access vlan
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with Switch FastEthernet0/3 (20).
% Incomplete command.
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (30), with Switch FastEthernet0/2 (1).
Switch(config-if)#switchport access vlan 99
% Access VLAN does not exist. Creating vlan 99
Switch(config-if)#
Copy
Paste
Top
Router3
PC-PT PC19
PC-PT PC20
PC-PT PC23
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ
Invite

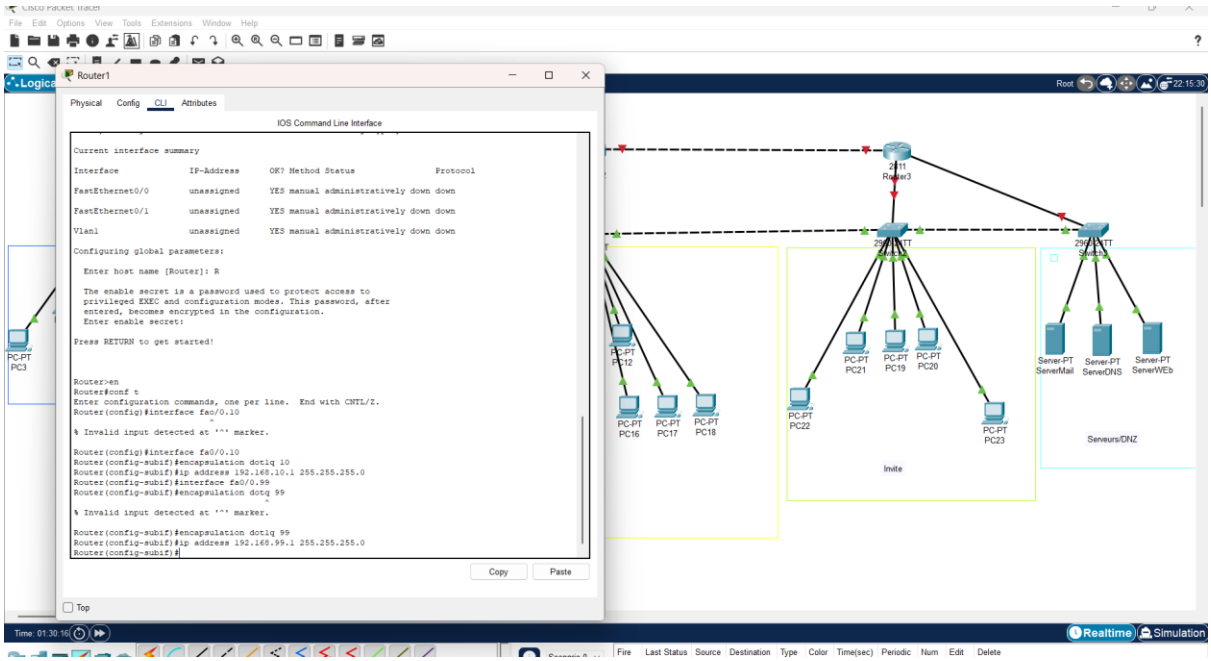

Cisco Packet Tracer
Logical
Physical
Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with Switch FastEthernet0/3 (30).
Switch(config-vlan)#name Invite
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DNZ
Switch(config-vlan)#interface range
% Invalid input detected at '^' marker.
Switch(config-vlan)#interface range fa0/1 -2
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with Switch FastEthernet0/3 (30).
0
Switch(config-if-range)#interface range fa0/1 -20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with Switch FastEthernet0/3 (30).
40
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#interface
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (40), with Switch FastEthernet0/2 (1).
% Invalid input detected at '^' marker.
Switch(config-if-range)#interface 0/24
% Invalid input detected at '^' marker.
Switch(config-if-range)#interface fa0/24
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (40), with Switch FastEthernet0/3 (30).
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (40), with Switch FastEthernet0/2 (1).
Copy
Paste
Top
Router3
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
PC-PT PC4
Administrations
PC-PT PC5
PC-PT PC8
PC-PT PC19
PC-PT PC20
PC-PT PC23
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ
Invite
Realtime
Simulation

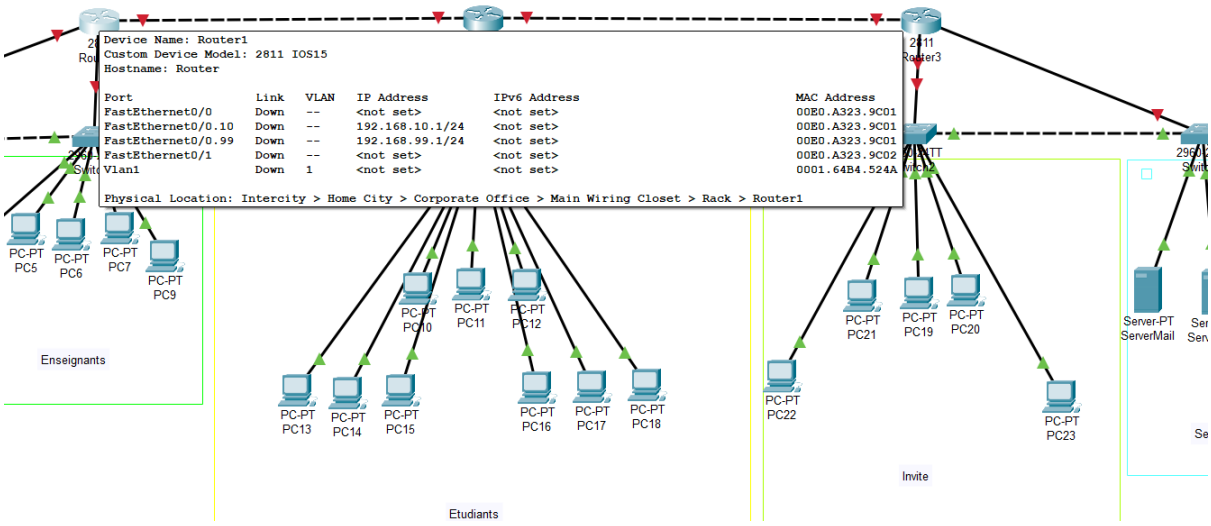


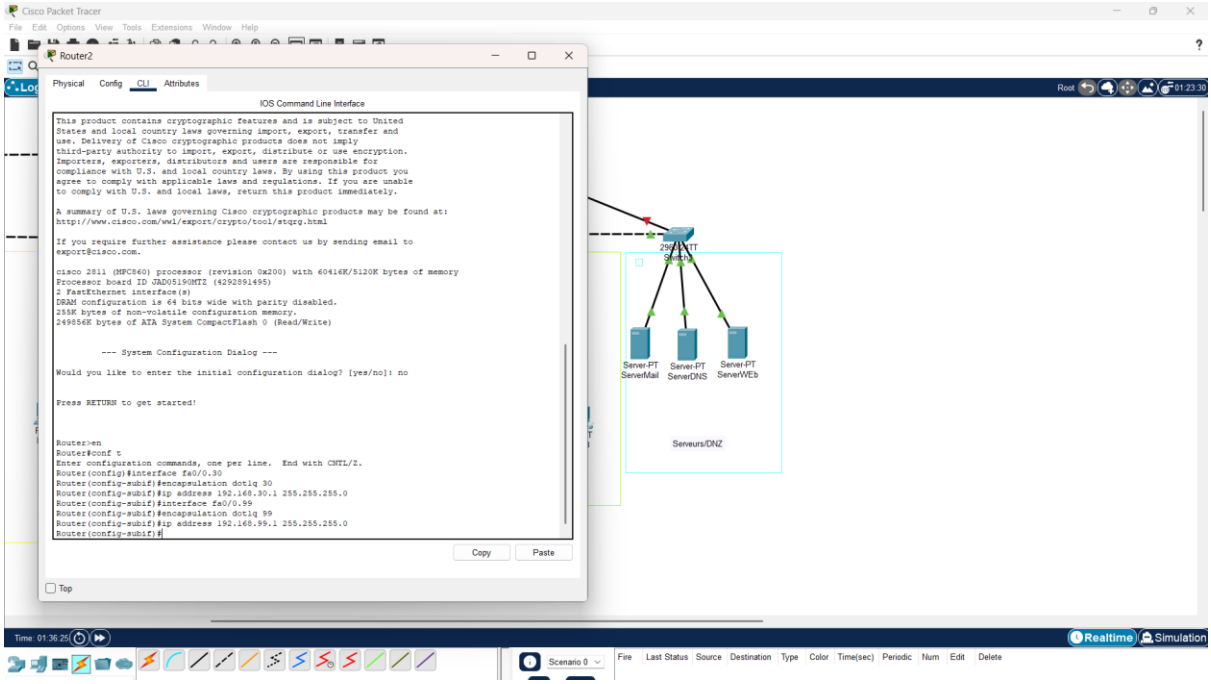
3. Configurer le routage inter-VLAN

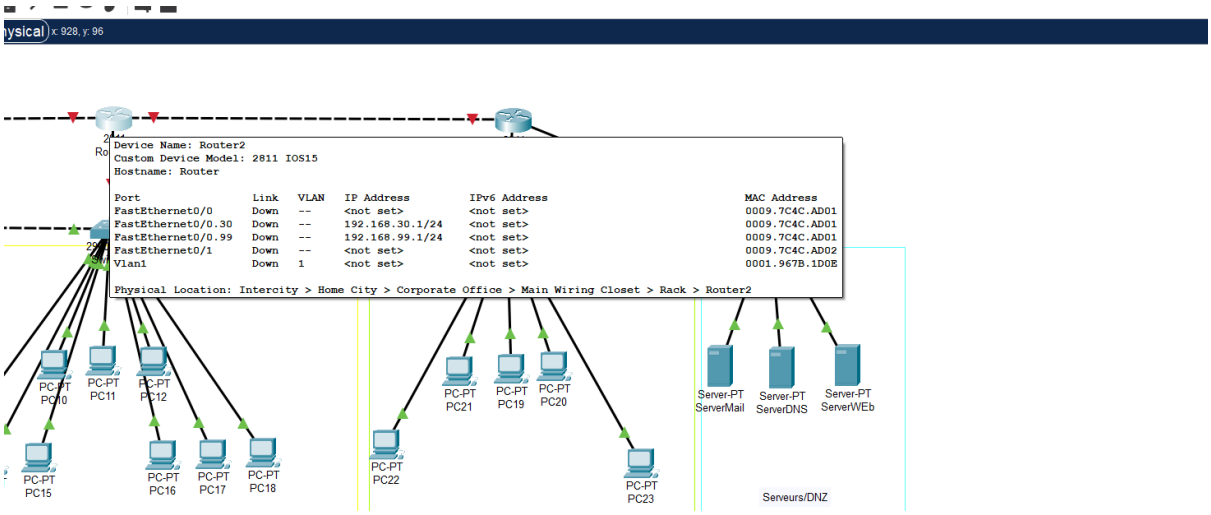
Sur les routeurs, créer des sous-interfaces VLAN avec des adresses IP pour permettre la communication entre VLANs autorisés.

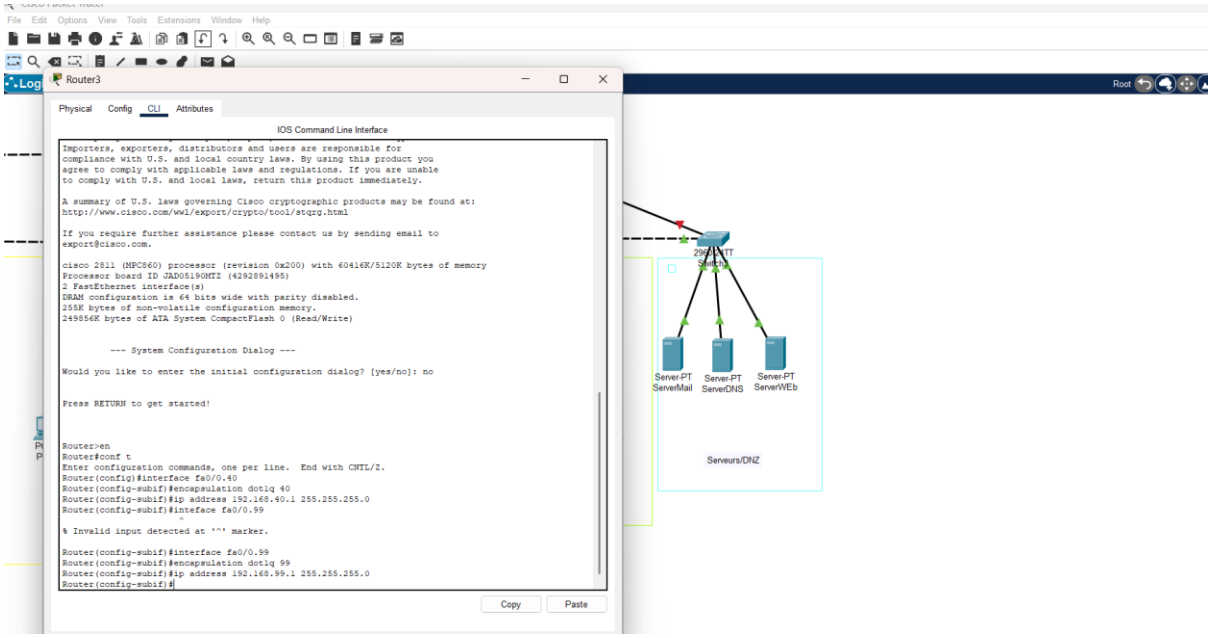
Vérifier la connectivité avec ping.

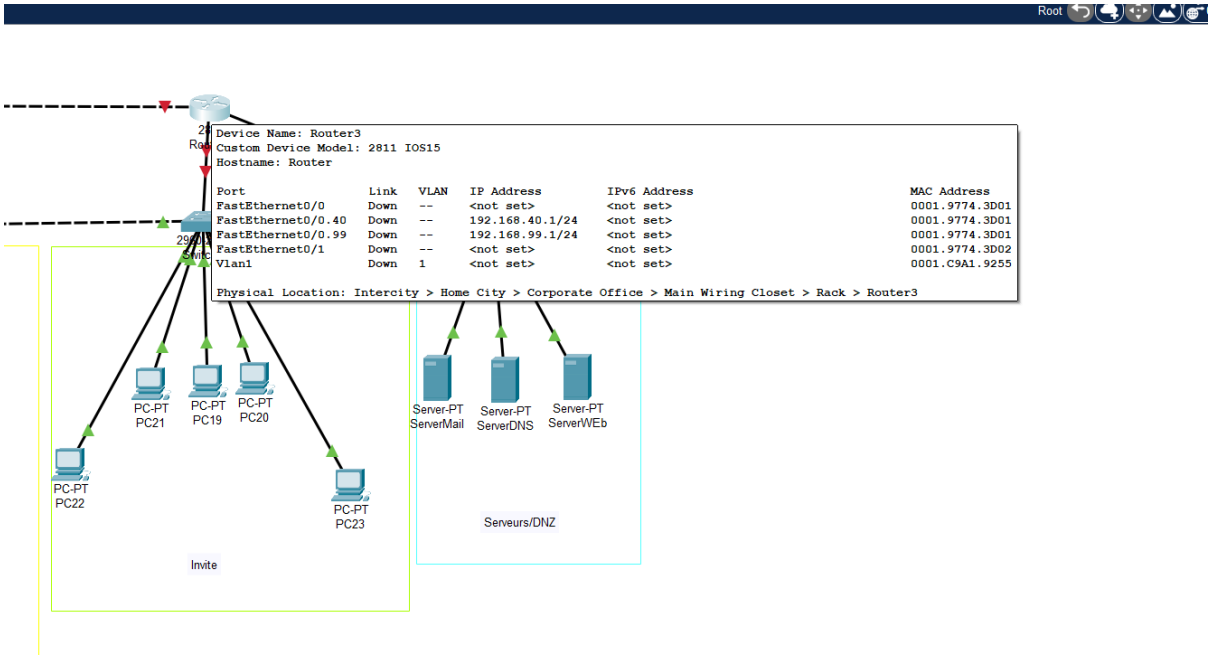


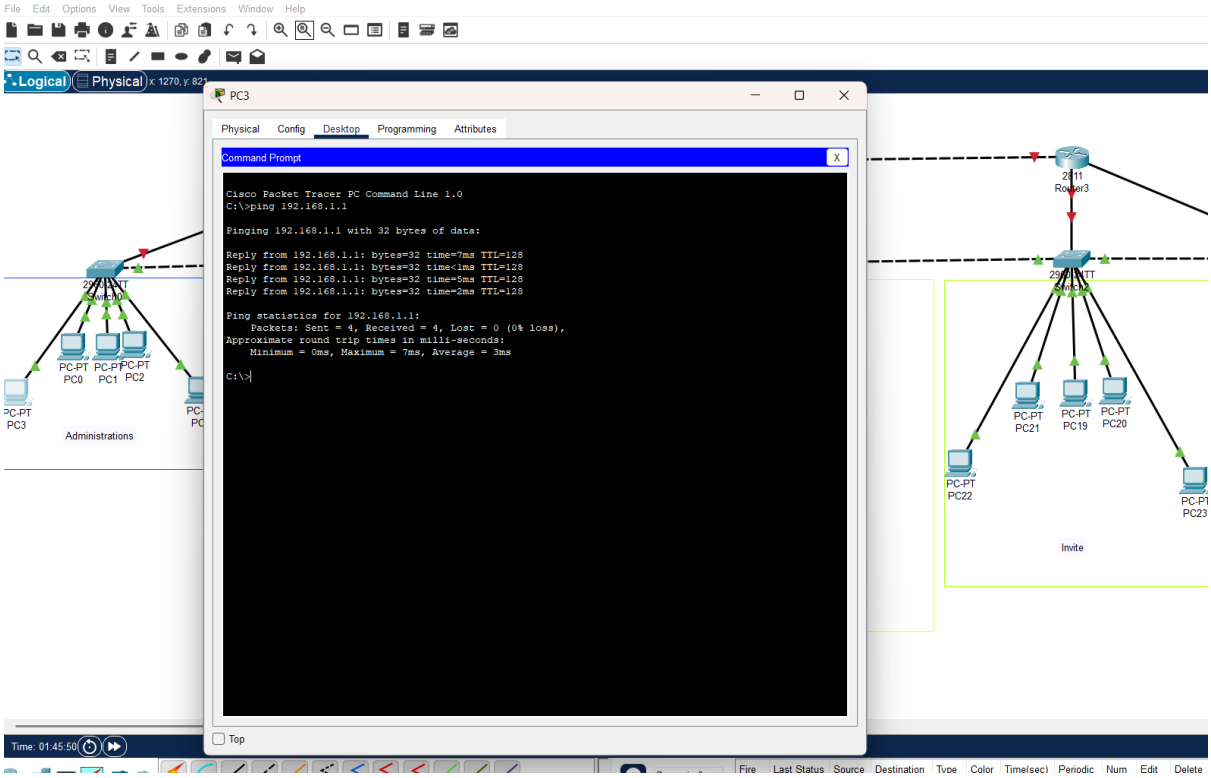

Device Name: Router1
Custom Device Model: 2811 IOS15
Hostname: Router
Port Link VLAN IP Address IPv6 Address MAC Address
FastEthernet0/0 Down -- <not set> <not set> 00E0.A323.9C01
FastEthernet0/0.10 Down -- 192.168.10.1/24 <not set> 00E0.A323.9C01
FastEthernet0/0.99 Down -- 192.168.99.1/24 <not set> 00E0.A323.9C01
FastEthernet0/1 Down -- <not set> <not set> 00E0.A323.9C02
Vlan1 Down 1 <not set> <not set> 0001.64B4.524A
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1
Router3
PC-PT PC5
PC-PT PC6
PC-PT PC7
PC-PT PC9
Enseignants
PC-PT PC10
PC-PT PC11
PC-PT PC12
PC-PT PC13
PC-PT PC14
PC-PT PC15
PC-PT PC16
PC-PT PC17
PC-PT PC18
Etudiants
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
PC-PT PC23
Invite
Server-PT ServerMail

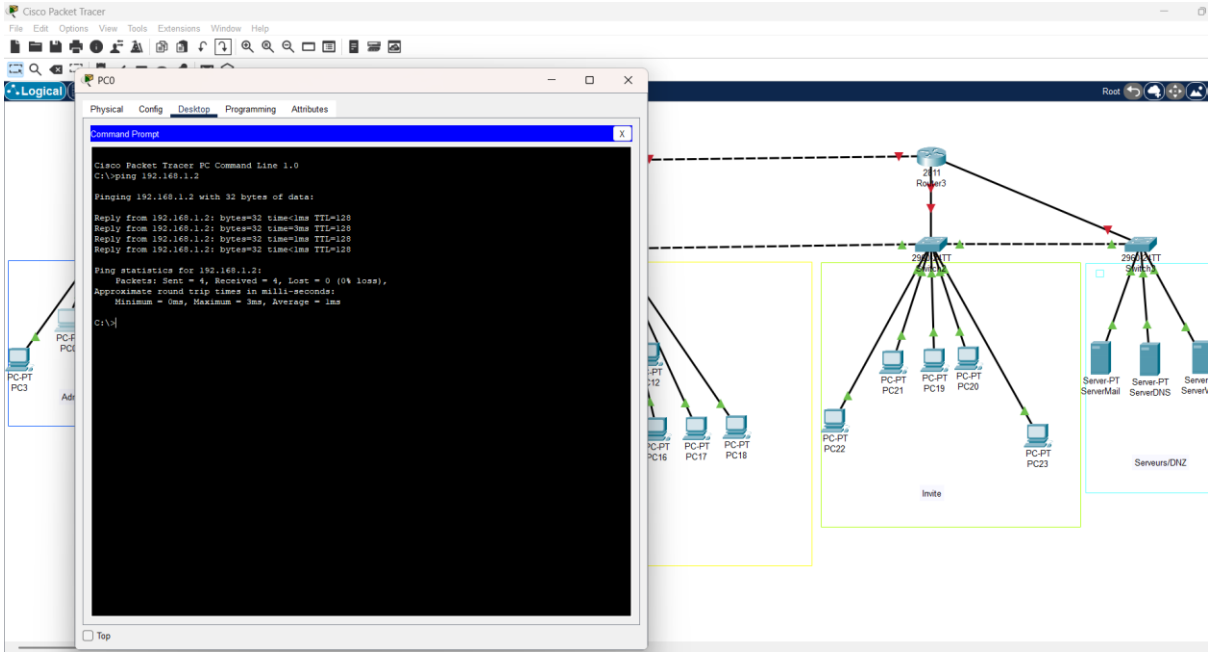

Cisco Packet Tracer
Router2
Physical Config CLI Attributes
IOS Command Line Interface
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0.30
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#interface fa0/0.99
Router(config-subif)#encapsulation dot1q 99
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#
Copy
Paste
Top
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ
Realtime
Simulation


x: 928, y: 96
Device Name: Router2
Custom Device Model: 2811 IOS15
Hostname: Router
Port Link VLAN IP Address IPv6 Address MAC Address
FastEthernet0/0 Down -- <not set> <not set> 0009.7C4C.AD01
FastEthernet0/0.30 Down -- 192.168.30.1/24 <not set> 0009.7C4C.AD01
FastEthernet0/0.99 Down -- 192.168.99.1/24 <not set> 0009.7C4C.AD01
FastEthernet0/1 Down -- <not set> <not set> 0009.7C4C.AD02
Vlan1 Down 1 <not set> <not set> 0001.967B.1D0E
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router2
PC-PT PC10
PC-PT PC11
PC-PT PC12
PC-PT PC15
PC-PT PC16
PC-PT PC17
PC-PT PC18
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
PC-PT PC23
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0.40
Router(config-subif)#encapsulation dot1q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#inteface fa0/0.99
% Invalid input detected at '^' marker.
Router(config-subif)#interface fa0/0.99
Router(config-subif)#encapsulation dot1q 99
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#
Copy
Paste
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ


Root
Device Name: Router3
Custom Device Model: 2811 IOS15
Hostname: Router
Port Link VLAN IP Address IPv6 Address MAC Address
FastEthernet0/0 Down -- <not set> <not set> 0001.9774.3D01
FastEthernet0/0.40 Down -- 192.168.40.1/24 <not set> 0001.9774.3D01
FastEthernet0/0.99 Down -- 192.168.99.1/24 <not set> 0001.9774.3D01
FastEthernet0/1 Down -- <not set> <not set> 0001.9774.3D02
Vlan1 Down 1 <not set> <not set> 0001.C9A1.9255
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router3
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
PC-PT PC23
Invite
Server-PT ServerMail
Server-PT ServerDNS
Server-PT ServerWEb
Serveurs/DNZ


File Edit Options View Tools Extensions Window Help
Logical
Physical
PC3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 3ms
C:\>
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Administrations
2811 Router3
PC-PT PC21
PC-PT PC19
PC-PT PC20
PC-PT PC22
Invite
Time: 01:45:50
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete



4. Mettre en place la sécurité réseau**

1. ACL (Access Control List)

Bloquer certains VLAN (ex. invités) d'accéder à la DMZ ou à l'administration.

Permettre aux VLAN autorisés de communiquer avec les serveurs.

2. Port Security

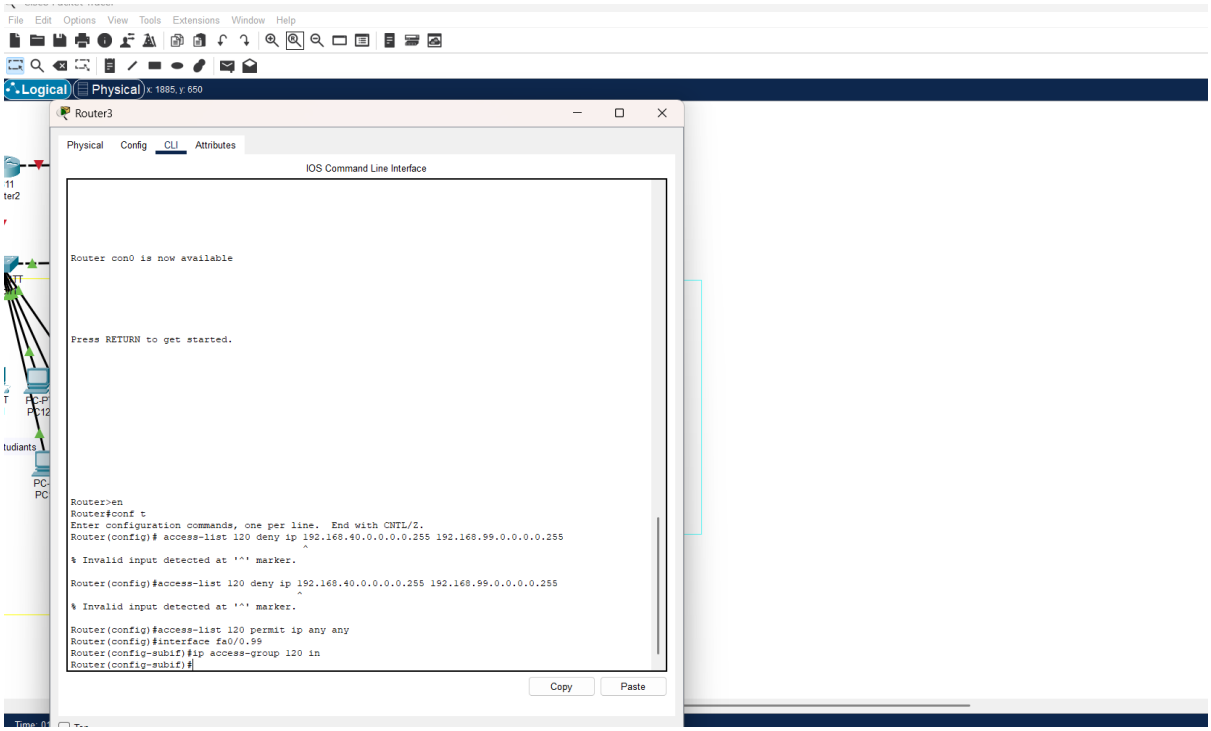
Limiter le nombre d'appareils par port sur les switches.

Configurer la violation pour restreindre l'accès si un PC non autorisé se connecte.

3. Sécurisation des routeurs

Désactiver Telnet, activer SSH pour l'administration.

Configurer des mots de passe forts.



Scénarios de test pour le TD

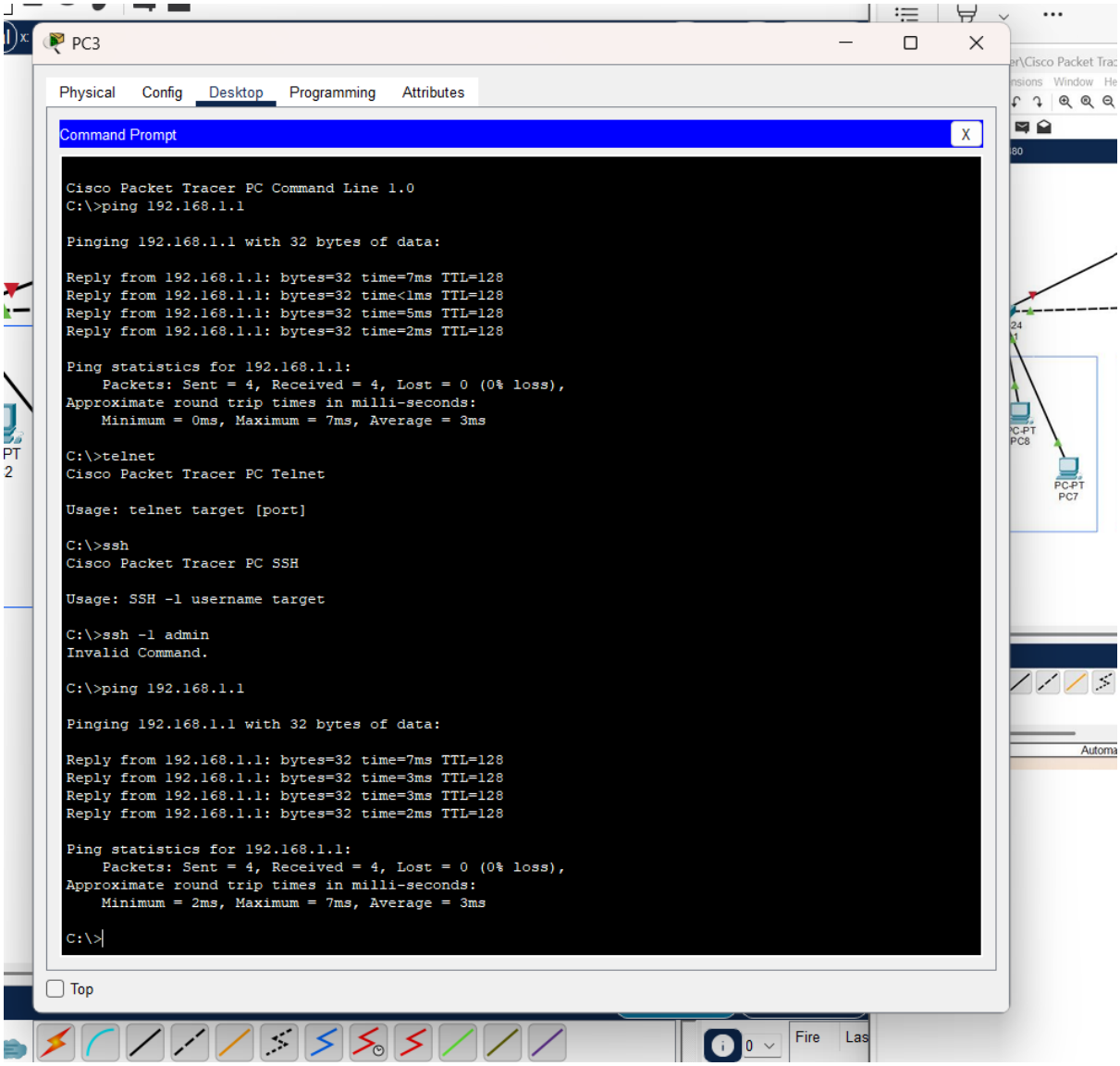
1. Ping entre VLAN autorisés : admin → enseignants → doit fonctionner.

2. Ping VLAN invités → serveur DMZ : doit échouer.

3. SSH routeurs : tester que Telnet est désactivé et SSH fonctionne.

4. Port Security : connecter un 3e PC sur un port limité → doit être restreint.

5. Accès aux serveurs : seuls admin et enseignants autorisés.


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 3ms
C:\>telnet
Cisco Packet Tracer PC Telnet
Usage: telnet target [port]
C:\>ssh
Cisco Packet Tracer PC SSH
Usage: SSH -l username target
C:\>ssh -l admin
Invalid Command.
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 7ms, Average = 3ms
C:\>
Top
PC-PT
PC7
Fire
Las

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>telnet
Cisco Packet Tracer PC Telnet

Usage: telnet target [port]

C:\>ssh
Cisco Packet Tracer PC SSH

Usage: SSH -l username target

C:\>ssh -l Invite
Invalid Command.

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=5ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\>
```

Ce TD m’a montré l’importance de la segmentation réseau avec les VLANs pour isoler les départements et renforcer la sécurité. Le routage inter-VLAN permet une communication contrôlée, tandis que l’usage de port-security, ACL et SSH protège l’accès et limite les menaces. En résumé, vous avez appris à bâtir une infrastructure plus organisée, sécurisée et résiliente face aux attaques.

Monsieur, je vous prie de bien vouloir excuser mon retard. J’ai rencontré des difficultés qui m’ont empêché de terminer le travail dans les délais impartis. Je vous présente mes sincères excuses.